FinFlux

Multilingual Financial Conversation Intelligence

Turning spoken finance into structured intelligence

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`Python 3.11+` • `FastAPI` • `React + Vite` • `Supabase pgvector` • `Groq LPU` • `AES-256`

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1. Executive Summary

Every day across India, millions of consequential financial decisions are made in conversation — over the phone, across the kitchen table, or in a bank branch. A family discusses prepaying a home loan. A first-generation investor asks about SIP adjustments in Hinglish. A financial advisor walks a client through a portfolio rebalancing in a mix of Hindi and English. These conversations carry real weight, yet they vanish the moment they end — unrecorded, unstructured, and unprotected.

FinFlux was built to change that. It is a production-grade, multilingual financial conversation intelligence platform that captures spoken financial discussions in Hindi, Hinglish, and English, securely transcribes them using AES-256 encryption, and runs them through a ten-model AI pipeline to produce executive-grade intelligence outputs in under seven seconds. The platform extracts actionable risk assessments, strategic intent, financial entities, and personalised reminders — all validated by a transparent, six-metric quality score so there are no black-box surprises.

What makes FinFlux genuinely different is its approach to language. Rather than forcing Hindi or Hinglish speech through an English translation layer, FinFlux processes conversations natively — preserving code-switching patterns, Devanagari script, and the bilingual financial lexicon that India's 800 million Hindi speakers actually use. Security is woven through every layer: from AES-256 audio encryption at rest to regex-based PII masking before any LLM sees the text, to prompt injection detection and row-level database isolation. The result is a full-stack, production-structured platform delivered within a hackathon timeframe — with a comprehensive REST API, twelve database migrations, and real-time financial keyword detection running live during recording.

2. Problem Statement

2.1 The Financial Inclusion Gap

Picture a scenario that plays out thousands of times each day across India. A family is discussing whether to prepay part of their home loan, switch their SIP amount, or take on a new personal loan. The conversation is urgent, real, and impactful — but it happens over tea, not on paper. It is spoken in Hinglish, not typed into an app. And when the call ends or the meeting breaks up, every commitment, every concern, and every decision evaporates.

This is not an edge case. In India, the vast majority of high-impact financial decisions occur in informal spoken conversations — between family members, with local advisors, or on the phone with bank representatives. These discussions involve critical commitments: EMI payments, loan prepayments, SIP adjustments, insurance renewals, and investment rebalancing. Despite their significance, **not a single one of these conversations is ever captured in a structured, searchable, or auditable format.**

2.2 Why This Gap Is Costly

The absence of structured financial conversation records creates a cascade of downstream problems:

- **Decision amnesia** — verbal financial commitments are forgotten within days, with no record to refer back to when a payment date arrives or a dispute emerges
- **Follow-up failure** — there is no mechanism to track stated intentions such as "I will increase my SIP next month" or flag unresolved decisions before they become problems
- **Risk blindness** — patterns of risky behaviour, such as over-leveraging or under-diversification, go undetected across a series of conversations that no one can see end-to-end
- **Language barrier** — virtually every existing fintech NLP tool is English-only, which excludes the 800 million+ Hindi and Hinglish speakers who make up the bulk of India's retail financial base
- **Privacy void** — sensitive financial data discussed verbally carries zero protection or traceability; there is no PII masking, no access control, no audit trail

Financial literacy in India is largely transmitted orally, not through documentation. Code-switching between Hindi, English, and Hinglish mid-sentence is the norm for urban and semi-urban users. No platform currently converts this multilingual speech into structured financial intelligence — FinFlux was built to be that platform.

3. Objectives

FinFlux was designed around ten engineering objectives, each directly addressing a dimension of the problem above:

1. **Multilingual ASR** — capture and transcribe financial conversations in Hindi, Hinglish, and English using Groq Whisper Turbo, with word-level timestamps
2. **Language-Aware Processing** — detect language composition with per-conversation Hindi/English/Hinglish percentage breakdowns and preserve code-switching patterns natively, never flattening them into a single language
3. **Financial NLP Intelligence** — extract structured financial entities such as loan amounts, EMI values, SIP rates, tenures, and interest rates using zero-shot NER across 18 financial entity types
4. **Sentiment & Risk Analysis** — classify financial sentiment through FinBERT's domain-specific model and compute multi-tier risk scores across four levels: CRITICAL, HIGH, MEDIUM, and LOW
5. **Expert-Level Reasoning** — deploy Qwen-32B for CFA-level analytical reasoning, surfacing strategic, technical, and compliance insights from each conversation
6. **Strategic Synthesis** — generate executive summaries, strategic intent classifications, and future insights using Llama-3.3-70B
7. **Semantic Memory via RAG** — enable cross-conversation retrieval using pgvector embeddings so the system can track financial decisions longitudinally over time
8. **Security-First Design** — enforce AES-256 audio encryption, PII masking before any LLM exposure, and prompt injection detection at every stage of the pipeline
9. **Transparent Quality Assurance** — provide real-time validation metrics with a six-component composite quality score, so every output comes with a confidence rating
10. **Production-Ready Platform** — deliver a full-stack FastAPI + React application with PDF/CSV report export, twelve database migrations, and thread-based conversation management

4. Target Users and Use Cases

4.1 Who FinFlux Serves

FinFlux is designed for anyone whose financial life happens in spoken conversations rather than typed inputs. That covers a wider range of users than most fintech products acknowledge:

Persona	Description	Key Need
Retail Investor	Individual managing a personal portfolio of SIPs, loans, and savings	Structured, searchable record of investment decisions
Rural / Hindi User	Hindi-speaking user with limited English proficiency who transacts in their native tongue	Native language support with no forced translation or transliteration
Financial Advisor	Professional managing a book of clients across multiple recorded conversations	Auditable conversation records with compliance-ready export
Family CFO	Household member coordinating EMIs, insurance, and savings verbally with family	Persistent structured records that prevent decision amnesia
Relationship Manager	Bank or NBFC officer in client-facing advisory meetings	Real-time topic capture and automated follow-up reminders

4.2 Three Real-World Scenarios

Scenario 1 — Code-Mixed Discussion (Hinglish):

"Yaar mera home loan ka EMI bahut zyada ho gaya hai, soch raha hoon prepayment kar doon, aur SIP bhi five thousand se badha ke eight thousand karna chahta hoon."

Without FinFlux, this conversation is gone the moment the call ends. With FinFlux, it becomes a structured intelligence record within 6.7 seconds: Topic = `DEBT_MANAGEMENT` (86% confidence), Risk = `MEDIUM`, Entities extracted = home loan, EMI, prepayment, SIP Rs. 5K → Rs. 8K, Strategic Intent = Consolidation, Quality Score = 0.794 (`GOOD`).

Scenario 2 — Rural Investor (Pure Hindi): A first-generation investor discusses mutual fund investments entirely in Hindi, using terms like *nivesh* and *kist* rather than "investment" and "EMI." FinFlux transcribes with Devanagari-aware processing, extracts entities from its bilingual financial lexicon, and generates a structured summary with follow-up reminders — no English required at any step.

Scenario 3 — Advisor Client Meeting: A financial advisor records a client call after a portfolio review. FinFlux produces an executive brief with risk classification, a timeline of extracted financial entities, and actionable insights — exportable as a compliance-ready PDF within seconds of the call ending.

5. Proposed Solution

5.1 High-Level Architecture

FinFlux is built on a **cloud-edge hybrid** philosophy: computation that demands speed and scale (ASR and large-language-model inference) runs on Groq's LPU cloud, while models that handle sensitive data (FinBERT, GLiNER, DeBERTa, XLM-RoBERTa) run locally, ensuring privacy and minimising latency. Ten specialist models are orchestrated into a coherent eight-stage pipeline, with each stage performing a well-defined role and handing a structured artefact to the next.

This is not a general-purpose chatbot bolted onto a transcription service. Every model in the stack was chosen for its specific domain strength: FinBERT because it was trained on financial corpora; GLiNER because its zero-shot NER architecture handles novel financial terminology without retraining;

XLM-RoBERTa because it natively understands 100+ languages including Hindi variants; and Qwen-32B because its instruction-following capability at CFA-analyst depth produces the kind of reasoning that financial professionals expect.

5.2 What Makes FinFlux Different

- **True Multilingual Pipeline** — Hindi, Hinglish, and English are processed natively with per-conversation language percentage breakdowns; a translation step is never inserted, so nuance is never lost
- **Ten-Model Expert Stack** — each model is a specialist with a single well-defined role; separation of concerns maximises accuracy at every stage and makes failures auditable
- **Security-First Architecture** — AES-256 encryption at rest, PII masking before any LLM exposure, and prompt injection detection ensure that sensitive financial data is protected at every handoff
- **Hybrid RAG Memory** — pgvector semantic retrieval across conversation history enables context-aware longitudinal analysis, so FinFlux remembers what a user said three months ago about their SIP target
- **Transparent Quality Scoring** — every output carries a six-metric composite quality score; there are no black-box results, and every confidence figure is traceable to a specific model output
- **Real-Time Detection** — a rolling five-second financial keyword detector runs during recording and gives the user live feedback on the topic being discussed before transcription even begins

6. System Architecture

6.1 Architecture Overview

FinFlux is a full-stack application composed of four tightly integrated layers: a **React + TypeScript frontend** that captures audio and presents results; a **FastAPI backend** that orchestrates the entire AI pipeline; an **NLP module pipeline** housing the ten specialist models; and a **Supabase PostgreSQL database** with pgvector for semantic memory and Row-Level Security for multi-tenant data isolation.

The diagram below shows the complete low-level architecture, illustrating how data flows from the browser's React component through the FastAPI API route, into the security module, through the NLP orchestration layer, and ultimately into the Supabase database tables with their associated caching and connection-pooling infrastructure.

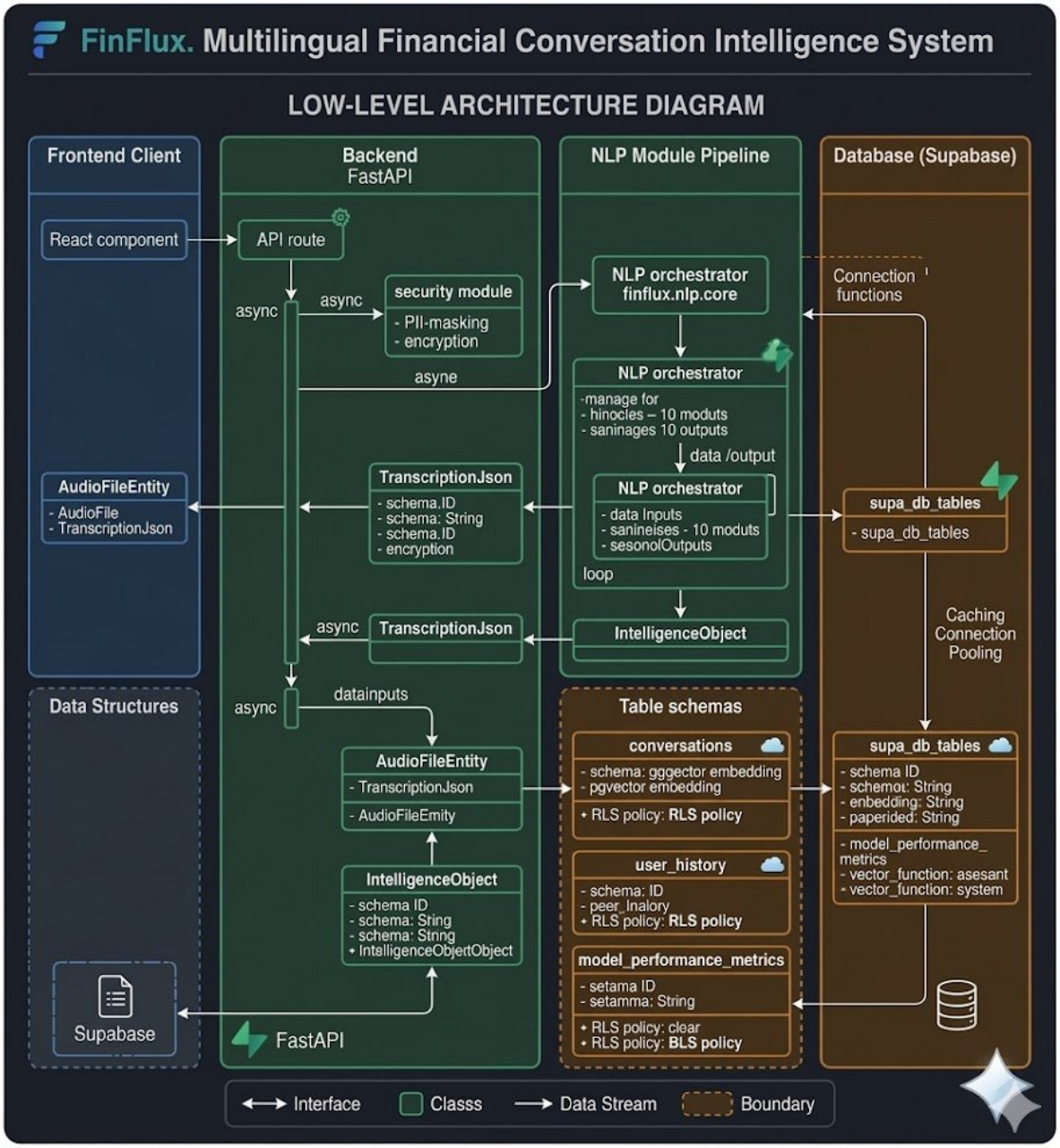


Figure 1: **FinFlux Low-Level Architecture Diagram.** The four-layer system: Frontend Client (React), Backend (FastAPI), NLP Module Pipeline, and Database (Supabase). Arrows indicate interface calls (*double-headed*), data streams (*right-pointing*), and asynchronous returns. Dashed boundaries denote module groupings.

Reading Figure 1 left to right:

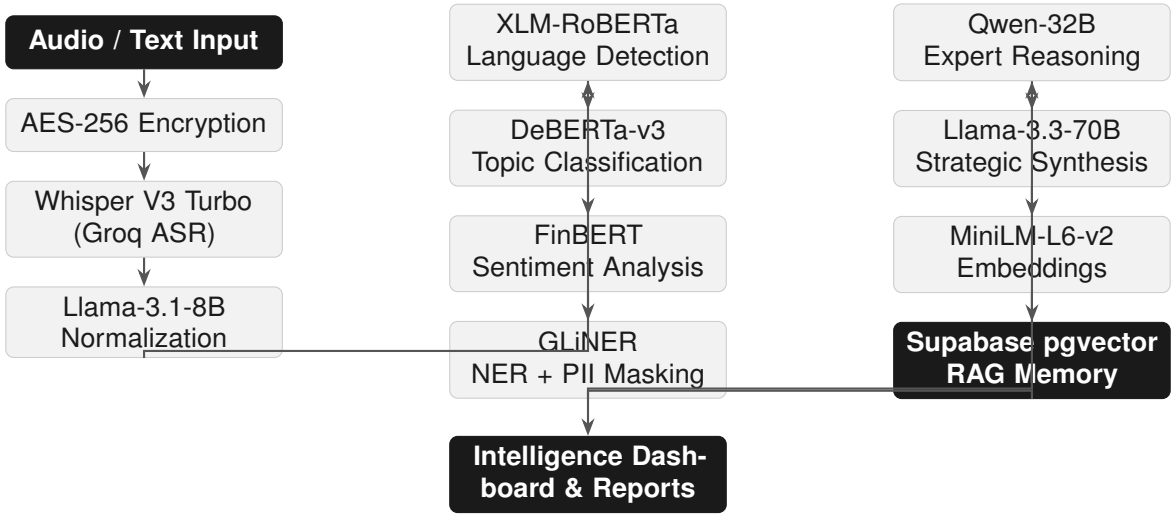
1. The **React component** (Frontend Client) sends audio or text to the FastAPI backend via an `async` API route call.
2. The **security module** immediately applies PII masking and AES-256 encryption before any data moves further.
3. The **NLP orchestrator** (`finflux.nlp.core`) manages the ten-model pipeline — routing inputs through transcription, normalisation, language detection, topic classification, sentiment analysis, entity extraction, expert reasoning, and embedding generation.
4. Results are persisted to **Supabase** (`supa_db_tables`) across three primary schema groups: `conversations` (with pgvector embeddings), `user_history`, and `model_performance_metrics` —

each protected by Row-Level Security policies.

5. An **IntelligenceObject** is returned asynchronously to the frontend along with the **TranscriptionJson**, which carries the schema ID, string fields, and encryption metadata.

6.2 Eight-Stage AI Pipeline

The pipeline block diagram below shows the sequential model stages. The first four stages run in a waterfall from audio capture through normalisation. Stages five and six (topic, sentiment, entity extraction) run concurrently where possible. Stages seven and eight (expert reasoning and embedding storage) close the pipeline and hand off to the dashboard.



6.3 Component Breakdown

#	Component	What It Does
1	Audio Capture & Encryption	Browser MediaRecorder captures 16 kHz PCM audio; Fernet AES-256 encrypts it at rest before any further processing
2	Multilingual ASR	Whisper V3 Turbo on Groq LPU transcribes Hindi, Hinglish, and English with word-level timestamps and mixed-script output
3	Transcript Normalization	Llama-3.1-8B cleans ASR artefacts (repeated words, filler hallucinations) while explicitly preserving Hinglish code-switching
4	Language Detection	XLM-RoBERTa classifies the dominant language and computes the Hindi/English/Hinglish percentage breakdown for each conversation
5	Topic & Sentiment	DeBERTa-v3 performs zero-shot classification into 15 financial categories; FinBERT simultaneously produces a sentiment probability distribution
6	Entity Extraction + Security	GLiNER extracts 18 financial entity types in zero-shot mode; all text is regex PII-masked before any LLM is invoked
7	Expert Reasoning & Synthesis	Qwen-32B generates CFA-level multi-wall analysis; Llama-3.3-70B synthesises the executive summary, strategic intent, risk level, and future insights
8	Storage & RAG Memory	MiniLM-L6-v2 generates 384-dim embeddings; results are persisted to Supabase bridge tables and pgvector for semantic retrieval

7. Methodology and Workflow

7.1 End-to-End Processing Flow

Understanding how a single conversation moves through FinFlux from recording to dashboard is important for appreciating the engineering decisions at each stage.

1. **User Input** — audio is captured via the Web Audio API at 16 kHz PCM mono, or the user pastes text directly. A rolling five-second financial keyword detector runs in parallel during recording, giving the user live topic indicators before transcription begins.
2. **Immediate Encryption** — the moment recording stops, the audio is encrypted using Fernet AES-256-CBC with PBKDF2HMAC key derivation (100 000 SHA-256 iterations) derived from `FINFLUX_SECRET`. A temporary WAV copy is sent to the ASR endpoint and then deleted; only the encrypted `.enc` file is retained.
3. **Transcription** — Groq Whisper V3 Turbo handles the multilingual transcription. Its training on diverse multilingual data means it handles mid-sentence code-switching and produces mixed-script output (Devanagari + Latin) natively. A local Whisper-small instance serves as a fallback if the Groq endpoint is unavailable.
4. **Normalisation** — Llama-3.1-8B receives the raw transcript and corrects ASR artefacts, repeated words, and filler hallucinations without translating or destroying the Hinglish patterns that make the conversation meaningful.
5. **Language Detection** — XLM-RoBERTa classifies the dominant language with a confidence score between 0.88 and 0.96, and computes a per-conversation Hindi/English/Hinglish percentage breakdown for reporting.
6. **Topic Classification & Sentiment** — DeBERTa-v3 zero-shot classifies the conversation into one of 15 financial categories (debt management, investment, insurance, and so on). FinBERT runs simultaneously and returns a probability distribution across positive, neutral, and negative financial sentiment.
7. **Entity Extraction & PII Masking** — GLiNER extracts 18 financial entity types in zero-shot mode. Crucially, all text is PII-masked at this stage, *before* any LLM receives it — PAN numbers, Aadhaar, phone numbers, and email addresses are replaced with labelled tokens.
8. **Expert Reasoning** — Qwen-32B generates a multi-wall expert analysis covering the strategic, technical, and compliance dimensions of the conversation. Llama-3.3-70B then synthesises an executive summary, classifies strategic intent, assigns a risk level, and surfaces future insights with urgency rankings.
9. **Embedding & Persistence** — MiniLM-L6-v2 generates a 384-dimensional dense embedding for the conversation. Results are persisted to Supabase bridge tables; the embedding is written asynchronously to pgvector to enable semantic retrieval in future sessions.
10. **Output** — the intelligence dashboard displays risk level, sentiment breakdown, extracted entities, strategic intent, quality score, and personalised reminders. PDF and CSV exports are available via the REST API without any additional steps.

7.2 Intent Routing

Not every message that arrives at FinFlux is a financial conversation recording. DeBERTa-v3 classifies every input before the full pipeline runs, routing it to one of three processing paths: (1) **Analysis route** — financial content detected; the full eight-stage pipeline executes; (2) **Financial inquiry** — a question about finance is detected; a direct LLM response is generated using RAG-retrieved context from prior conversations; (3) **General route** — non-financial content is detected; a lightweight casual response is returned without invoking the heavy pipeline.

8. Technologies Used

Layer	Technologies
AI / ML	Groq API (Whisper V3 Turbo, Llama-3.1-8B, Qwen-32B, Llama-3.3-70B); HuggingFace (ProsusAI/finbert, nli-deberta-v3-small, papluca/xlm-roberta-base, gliner_medium-v2.1); Sentence-Transformers (all-MiniLM-L6-v2)
Frameworks	PyTorch, HuggingFace Transformers, PEFT, Accelerate
Backend	FastAPI + Uvicorn (async ASGI); Supabase PostgreSQL + pgvector; SQLAlchemy ORM + SQLite fallback; Supabase Auth (JWT); 12 SQL migrations
Security	Fernet AES-256-CBC; PBKDF2HMAC; regex PII masking; Supabase Row-Level Security
Frontend	React 18 + TypeScript + Vite; Web Audio API (16 kHz PCM); Web Speech API; Lucide React
Reporting	ReportLab (PDF); Pandas (CSV)

9. Multilingual Handling

9.1 Language Detection Architecture

The first thing FinFlux establishes about any conversation is its linguistic identity. `papluca/xlm-roberta-base-lang` serves as the primary classifier, providing: (a) the dominant language with a confidence score typically between 0.88 and 0.96; (b) a per-conversation Hindi/English/Hinglish percentage breakdown for reporting and analytics; and (c) Devanagari script preservation throughout the full downstream pipeline, so Hindi words never silently get transliterated into Latin characters.

9.2 Code-Switching and Hinglish

Hinglish is not a degraded form of Hindi or a simplified form of English — it is the native register of financial communication for hundreds of millions of urban and semi-urban Indians, and it must be handled as a first-class input. FinFlux does this through four interlocking mechanisms:

- **Native ASR** — Whisper Turbo natively handles code-switched speech and produces mixed-script transcripts without requiring language-switching prompts
- **Artefact-preserving normalisation** — Llama-3.1-8B corrects transcription errors without translating or collapsing the Hinglish patterns that distinguish *"EMI bahut zyada ho gaya"* from any pure-English equivalent
- **Bilingual financial lexicon** — a 150+ term lexicon covering both English financial terms (EMI, SIP, CAGR) and their Hindi Devanagari equivalents (*kist*, *nivesh*, *byaj*) enables accurate entity extraction across both scripts
- **No forced translation** — text is processed throughout the pipeline in its original language; translation is never inserted as an intermediate step

9.3 Multilingual Challenges and Solutions

Challenge	Solution
Code-switching mid-sentence	Whisper Turbo handles mixed-script speech natively; Llama normalisation preserves switching patterns rather than forcing a single script
Hindi financial terminology	150+ term bilingual lexicon with Devanagari and Latin variants covers the vocabulary gap between formal financial English and spoken Hindi
Accent and dialect diversity	Groq Whisper trained on diverse multilingual data generalises across Indian regional accent variations without per-accent fine-tuning
Devanagari in NER and metrics	Unicode-aware tokenisation and fuzzy matching ensure that Devanagari tokens are not lost or misaligned during entity extraction

10. Data Handling and Processing

10.1 Input Data Types

FinFlux accepts two input modalities: audio (WAV at 16 kHz, 16-bit PCM, mono) captured via the Web Audio API and stored immediately as AES-256 encrypted `.enc` files; and direct text input in Hinglish, Hindi, or English, which enters the NLP pipeline at the normalisation stage, bypassing the ASR step.

10.2 Data Flow

The data lifecycle inside FinFlux is designed to minimise the window during which sensitive data is unprotected:

1. Raw audio is encrypted to `.enc` immediately; the temporary WAV copy is deleted as soon as transcription is complete
2. The transcript is PII-masked before reaching any LLM; no model ever sees a raw PAN number, Aadhaar, or phone number
3. Analysis results are persisted to Supabase bridge tables as the primary store, with SQLite as a local fallback if the cloud connection is unavailable
4. Embeddings (384-dim dense vectors) are written asynchronously to pgvector to avoid adding latency to the main response path

10.3 Database Schema

Table	Purpose
`ai_conversation_threads`	Thread-level grouping with per-user isolation enforced by Row-Level Security
`ai_conversation_messages`	Per-turn message storage for both user and assistant roles
`ai_message_embeddings`	384-dim pgvector embeddings enabling semantic retrieval across conversations
`ai_conversation_quality_metr`	Per-conversation quality scores and model version tags for auditability

11. AI Model Details

The ten models in the FinFlux stack were each selected for a specific domain strength, not for general capability. The table below documents each model's role, inference location, and key technical characteristics.

Model	Task	Key Capability	Inference
Whisper V3 Turbo	ASR	Multilingual STT with word-level timestamps and mixed-script output	Groq Cloud
Llama-3.1-8B	Normalize	Artefact removal while preserving code-switching patterns	Groq Cloud
XLM-RoBERTa	Lang Det.	20+ language classes with per-class confidence scoring	Local GPU
DeBERTa-v3-small	Topic	Zero-shot classification across 15 financial categories	Local GPU
FinBERT	Sentiment	Financial domain-specific Pos / Neu / Neg probability scoring	Local GPU
GLiNER Medium v2.1	NER	Zero-shot extraction across 18 financial entity types	Local GPU
Qwen-32B	Reasoning	CFA-level analytical reasoning across strategic, technical, and compliance dimensions	Groq Cloud
Llama-3.3-70B	Synthesis	Executive summary generation with strategic intent and risk classification	Groq Cloud
all-MiniLM-L6-v2	Embeddings	384-dim dense vectors optimised for semantic similarity search	Local CPU
Supabase pgvector	Retrieval	Cosine similarity search across conversation history with metadata filtering	Supabase

All models operate in zero-shot or instruction-following mode within the current hackathon implementation. No model was fine-tuned from scratch for this submission. However, LoRA fine-tuning of Whisper on an Indian financial speech corpus is planned as a direct post-hackathon step — the pipeline assets and training configuration are already in place.

12. Features

Every feature in the table below is fully implemented in the current FinFlux build:

Feature	Status
Secure voice capture with real-time waveform visualisation	**Implemented**
Multilingual support (Hindi, Hinglish, English) with language percentage breakdown	**Implemented**
10-model AI pipeline with expert reasoning (Qwen Wall of Logic)	**Implemented**
Financial sentiment analysis with FinBERT probability breakdown	**Implemented**
Zero-shot financial NER (18 entity types via GLiNER)	**Implemented**
Four-tier risk classification (CRITICAL / HIGH / MEDIUM / LOW)	**Implemented**
Hybrid RAG with pgvector semantic search and metadata filtering	**Implemented**
AES-256 audio encryption + PII masking (PAN / Aadhaar / Phone / Email)	**Implemented**
Prompt injection detection with 20+ keyword pattern matching	**Implemented**
Personalised financial reminders generated from conversation patterns	**Implemented**
Future insights generation with urgency ranking	**Implemented**
Quality validation dashboard with six-metric composite scoring	**Implemented**
PDF and CSV report export in executive format	**Implemented**
Real-time financial topic detection during active recording	**Implemented**
Thread-based chat with persistent conversation history	**Implemented**
Transcript editing with re-analysis trigger	**Implemented**

13. Results and Output

13.1 Sample Pipeline Output

The following table shows the complete FinFlux output for the Hinglish home-loan/SIP conversation from Section 4. The full pipeline — from audio submission to intelligence dashboard — completed in 6.7 seconds.

Field	Value
Financial Topic	`DEBT_MANAGEMENT` (86% confidence)
Risk Level	`MEDIUM`
Sentiment	Neutral dominant — Positive 19%, Neutral 54%, Negative 27%
Strategic Intent	Consolidation
Language Mix	Hindi 40.1%, Hinglish 34.8%, English 25.1%
Entities Extracted	home loan (95%), EMI (92%), prepayment (90%), SIP Rs. 5K → Rs. 8K
Quality Score	0.794 — **GOOD** tier
Processing Time	6.7 seconds end-to-end

13.2 Quality Scoring Framework

Every FinFlux output carries a composite quality score Q computed as a weighted sum of six model-derived metrics. The weights reflect the relative importance of each component to overall output reliability:

$$Q = 0.30 \cdot C_{\text{ASR}} + 0.20 \cdot \frac{N_{\text{NER}}}{100} + 0.20 \cdot R_{\text{ROUGE}} + 0.15 \cdot \frac{E_{\text{align}}}{100} + 0.10 \cdot C_{\text{lang}} + 0.05 \cdot F_{\text{rel}}$$

ASR confidence carries the highest weight (0.30) because transcription quality is the single largest determinant of downstream accuracy. NER coverage and ROUGE recall each contribute 0.20,

reflecting the importance of entity completeness and summary fidelity. Entity alignment, language confidence, and financial relevance round out the score with decreasing weights.

Metric	Description	Typical Range
ASR Confidence	Average word-level probability from Whisper	0.87 – 0.94
NER Coverage	Financial lexicon term match rate in extraction	65 – 80%
ROUGE-1 Recall	Summary vs. transcript token overlap	0.55 – 0.72
Entity Alignment	Entities present in summary (fuzzy match)	70 – 85%
Language Confidence	XLM-RoBERTa per-class confidence	0.88 – 0.96
Financial Relevance	DeBERTa top-topic confidence score	0.72 – 0.91

Quality tiers: `EXCELLENT` ($Q > 0.82$) `GOOD` ($0.65 < Q \leq 0.82$) `ACCEPTABLE` ($0.45 < Q \leq 0.65$) `LOW` ($Q \leq 0.45$)

13.3 REST API Surface

Method	Endpoint	Description
`POST`	`/api/analyze`	Submit audio or text through the full intelligence pipeline
`POST`	`/api/chat`	Text-based chat with intent routing and RAG context
`GET`	`/api/results`	Retrieve the authenticated user's conversation history
`GET`	`/api/quality/summary`	Aggregate quality dashboard across all conversations
`GET`	`/api/report/{id}`	Export conversation report as PDF or CSV
`POST`	`/api/search/semantic`	Semantic memory search across pgvector embeddings
`GET`	`/api/health`	Check model and service status
`DELETE`	`/api/threads/{id}`	Delete a conversation thread and purge its history

14. Challenges Faced

Building FinFlux within a hackathon timeframe while maintaining production-grade standards required navigating several non-trivial engineering challenges:

Challenge	Solution
Hinglish ambiguity confuses standard NER	GLiNER's zero-shot architecture handles novel Hinglish financial patterns without retraining, using financial entity labels as the only supervision signal
Indian financial terms span Devanagari and Latin scripts	A 150+ term bilingual lexicon covers both scripts, enabling entity extraction to work regardless of whether the user types *kist* or EMI
ASR hallucinations on accented speech	Llama-3.1-8B normalisation corrects artefacts post-transcription; a local Whisper-small fallback handles edge cases where Groq cloud is unavailable
Latency with 10 sequential models	Groq LPU delivers sub-second inference for cloud models; lightweight local models run concurrently where the pipeline allows, keeping total latency under 7 seconds
LLM context window constraints	Thread-level context is limited to the last four turns, supplemented by semantically retrieved prior context from pgvector
Multi-tenant data isolation	Supabase RLS with `user_id`-gated policies; backend bridge tables (`ai_*`) deny all direct access from `anon` and `authenticated` roles

15. Future Scope

The current FinFlux implementation is a production-structured foundation. The following roadmap reflects the most impactful directions for extending it:

1. **Additional Languages** — expand native support to Tamil, Telugu, Kannada, Marathi, and Bengali, each with a language-specific financial lexicon, to reach the full breadth of India's linguistic diversity
2. **Real-Time Streaming ASR** — replace the record-then-transcribe workflow with WebSocket-based streaming transcription for live, in-conversation intelligence
3. **Banking API Integration** — connect with NPCI/UPI rails, bank statement parsers, and credit bureau APIs to enrich extracted entities with verified transactional data
4. **LoRA Fine-Tuning** — fine-tune Whisper on a curated Indian financial speech corpus using LoRA adapters; the training pipeline and dataset curation tools are already in place from the hackathon build
5. **Mobile Application** — a React Native client for on-the-go financial conversation capture, extending FinFlux to the 600+ million smartphone users in India
6. **Automated Financial Plans** — generate structured financial action plans from longitudinal conversation history, moving FinFlux from intelligence to active guidance
7. **Compliance Module** — integrate SEBI and RBI regulatory compliance checks on detected financial advice, enabling use in regulated advisory contexts
8. **Voice Biometrics** — add speaker identification for multi-party financial meetings, enabling entity attribution at the speaker level rather than the conversation level

16. Security and Privacy

Security in FinFlux is not a feature — it is a constraint that shapes every design decision. Financial conversations contain some of the most sensitive personal data that exists: account numbers, loan details, PAN cards, Aadhaar identifiers, and candid disclosures about personal financial stress. Every layer of FinFlux is designed to protect that data.

16.1 Audio Encryption

All recorded audio is encrypted at rest using **Fernet AES-256-CBC** with PBKDF2HMAC key derivation (SHA-256, 100 000 iterations). The encryption key is derived from the `FINFLUX_SECRET` environment variable and never stored in plaintext. Encrypted `.enc` files are stored in `data/encrypted_audio/` and are decrypted only transiently for ASR processing. The temporary decrypted WAV is deleted immediately after the transcription call completes.

16.2 PII Masking

All transcribed text passes through a regex PII masking stage before reaching any LLM. The masking covers five PII categories:

PII Type	Detection Pattern	Replacement Token
PAN Card	5 uppercase + 4 digits + 1 uppercase	`[MASKED_PAN]`
Aadhaar	12 consecutive digits or 4-4-4 format	`[MASKED_AADHAAR]`
Phone	Indian mobile (6–9 prefix, 10 digits)	`[MASKED_PHONE]`
Email	Standard RFC 5322 email regex	`[MASKED_EMAIL]`
Bank Account	9–18 digit numeric sequences	`[MASKED_ACCOUNT]`

16.3 Access Controls

- **Prompt injection detection** — 20+ keyword patterns scan every incoming message for override phrases, persona manipulation attempts, and financial advice solicitation; detection results are included in response metadata so they are auditable
- **Row-level security** — Supabase RLS with `user_id`-based policies is active on every table; backend bridge tables (`ai_*`) deny all direct access from `anon` and `authenticated` roles; data flows exclusively through audited API endpoints
- **Compliance readiness** — the architecture is designed for future alignment with India's Digital Personal Data Protection (DPDP) Act; a user history purge endpoint (`POST /api/history/purge`) is already implemented and accessible

17. References

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18. Conclusion

FinFlux set out to answer a specific question: can multilingual financial conversation intelligence be delivered at production quality within a hackathon timeframe? The answer, demonstrated by this submission, is yes — and the implications go well beyond the competition.

India's financial inclusion challenge is not primarily a product problem or an infrastructure problem. It is a language problem. The tools that exist to help people make better financial decisions were built for English speakers, and the 800 million Hindi and Hinglish speakers who dominate India's retail financial landscape have been excluded from the benefits of intelligent financial tooling as a result. FinFlux is the first platform to natively process this population's actual speech — code-switched, accented, bilingual, and informal — and convert it into structured, auditable intelligence.

The platform delivers impact across three measurable dimensions. **Language inclusion**: native Hindi, Hinglish, and English processing without forced translation extends intelligent financial tooling to hundreds of millions of speakers previously excluded. **Decision capture**: ephemeral spoken financial commitments are converted into persistent, searchable, structured records with full provenance, ending the cycle of decision amnesia that currently affects informal financial planning. **Security**: end-to-end protection via AES-256 encryption, PII masking, prompt injection detection, and row-level tenant isolation ensures that the most sensitive personal data — financial conversations — is handled responsibly at every stage.

FinFlux is not a minimum viable prototype. It is a production-structured platform with a full-stack implementation, a comprehensive REST API across eight endpoints, twelve database migrations, thread-based conversation management, PDF and CSV export, and real-time financial keyword detection. The architecture is designed to scale from this hackathon submission to an enterprise deployment serving India's financial inclusion challenge at national scale — with the roadmap, the pipeline assets, and the engineering foundation already in place to do so.